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3D Graphene-Based Adsorbents: Synthesis, and Spectroscopic Analysis for Oil Field Applications

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This study explores the use of 3D graphene and its derivatives as an adsorbent for wastewater contaminants like heavy metals, dyes, and oil, focusing on their effectiveness in creating environmentally friendly materials. Recent advancements in graphene synthesis have made it possible to create 3D porous graphene aerogel (GA) structures that preserve the distinctive qualities of single-layer and multilayer graphene. Typical synthesis of GA involves reducing graphene oxide in solution to form a 3D reduced graphene hydrogel network. The aerogel is then formed by freeze-drying the solvent out of the pores and swapping it with air. Using high-temperature annealing can significantly reduce the quantity of flaws in the freeze-dried GA. GAs are synthetic materials that are lightweight, elastic, highly conductive, and structurally strong that have been explored for various applications. Compared with traditional 2D graphene materials, GAs provide additional values, such as high porosity, large effective surface area, and robustness, which makes them promising for various macroscopic applications and the development of hybrid graphene-based adsorbents. Temperature-dependent assembly of 3D graphene networks in free-standing condition of graphene aerogels that were covalently cross-linked. To elucidate the function of defects in the materials, a controlled transition between reduced-graphene oxide and graphene exploiting high-temperature annealing was studied. Raman spectra reveal two intense peaks in non-annealed samples, while annealed samples at high temperature show an additional 2D peak and a reduced D peak intensity. This demonstrates that the annealing was efficient in improving the crystallinity and removing oxygen functional groups from the GA, facilitating a transition from reduced graphene oxide to graphene. Oxygen functional groups are unstable and desorb with higher annealed temperature: resulted higher hydrophobic and become more oleophilic as observed from the experiment. This outlines future prospects and viewpoints while suggesting directions and suggestions for more study to comprehend oil removal processes from wastewater.

Introduction

Energy demands rise with rapid industrialization, and wastewater quality deteriorates as a result of industrial wastes. Due to a number of factors, such as the discharge of heavy machinery and chemicals that thrive in water, this presents a serious environmental risk. The crude oil industry generates wastewater containing 60-3000 mg/L of oil and grease, along with chemical pollutants like organic solvents and metal ions.

Large volumes of oily wastewater with periodic oil disposal may cause ecological imbalance.¹⁻⁵ This pollution components involved in wastewater are complex, causing harm to human health and ecological balance. High-performance water purification requires powerful sample preparation strategies, relying on advanced adsorbent materials for high clean-up and enrichment efficiency. Low-dimensional materials like nanoparticles, colloidal nanomaterials, carbon nanotubes, and 2-D nanosheets have shown exceptional performance in purifying polluted water due to their high specific surface areas and selectivity.⁶⁻¹² Oil and grease can be removed from industrial effluent using a variety of treatment techniques, such as physical, chemical, electrical, and biological approaches.^{13, 14} These strategies need to be assessed in light of factors such growing wastewater loads, costs, and the creation of any other hazardous waste. Adsorption's simplicity, great efficacy, and convenience of use make it a viable technique in this regard. However, it is not feasible due to the high operational costs. Because it is less expensive to produce, graphene, a carbon nanomaterial, is frequently reported in wastewater cleanup.^{1-3, 5} Because of its exceptional optical and electrical characteristics, including high electron mobility, charge-density tunability, and wideband absorption, graphene has become a remarkable two-dimensional (2D) material in compared with other 2D materials.^{15, 16} In comparison to graphene, graphene oxide (GO), a byproduct of graphene sheet oxidation, has inferior mechanical and electrical properties.¹⁷ The hydrophilicity of GO is enhanced by oxygen-containing functional groups, which results in a stable colloidal suspension in aqueous conditions. GO is a viable material for adsorption and surface reaction applications because to its large surface area, superior colloidal stability, and high aspect ratio for modification.^{17, 18}

This 2D graphene-based nanomaterials consider as one of the good adsorbents for this purpose. Gathering low-dimensional nanostructured materials into three-dimensional macrostructures is one likely technique to address the issues of water pollution. With different frameworks, researchers have created 3D graphene-based adsorbents that retain their inherent qualities while acquiring new collective physiochemical properties.⁴ A new era in sample preparation is made possible by these 3D Graphene-based adsorbents enhanced mass transport and surface accessibility, as well as their ability to be altered with useful parts. 3D graphene networks formed in free-standing covalently cross-linked graphene structures that preserve the distinctive qualities of single-layer/multilayer graphene have been made possible in recent years by developments in graphene production.^{19, 20} GAs have a huge effective surface area and high porosity when compared to conventional 2D graphene. They are also lightweight, elastic, highly conductive, and mechanically resilient, which makes them interesting for a variety of applications, such as strain and pressure sensing.²¹

By enhancing the crystallinity and eradicating oxygen functional groups from the GA, the annealing procedure successfully facilitated the transition from reduced graphene oxide to graphene, as demonstrated in this work. Oxygen functional groups are unstable and desorb with higher annealed temperature: resulted higher hydrophobic and become more oleophilic as observed from the experiment. This proposes avenues and recommendations for ongoing research to understand oil elimination procedures from wastewater, outlining future prospects and perspectives.

Result & Discussion

The detail synthesis procedure of 3D Graphene discussed elsewhere.¹⁹ Briefly 60 millilitres of deionized water were combined with 120 milligrams of graphene oxide (GO) powder. After being carefully mixed, the GO solution was submerged in an ultrasonic bath (sonication for 1 hour at 45°C). A Teflon-lined hydrothermal stainless-steel autoclave was filled with 100 ml of the GO mixture, which was then annealed for six hours at 200 °C. A chemically reduced and self-assembled three-dimensional hydrogel structure was produced following the hydrothermal procedure. To obtain a cylindrical sample of stable reduced graphene oxide (rGO) aerogel, the hydrogel was repeatedly rinsed with deionized water to get rid of any leftovers, and then freeze-dried in a vacuum for 16 hours at -70 °C. After being stored at room temperature, the resulting

rGO aerogel was vacuum annealed for 30 minutes at room temp and at high temp 2700 °C. The morphology of this engineered cross-linked graphene networks in GA with the same shape but varying degrees of structural flaws analysed by Scanning electron microscopy and Raman spectroscopy investigation as shown in Figure 1. The GA shows a 3D porous network that is highly linked (Fig. 1A, B). A wide range of sizes, around 1 to 3-micrometers diameter size pores can see in the SEM image. Graphene sheets are layered in extremely thin thicknesses to form the walls of the pores. The D and G bands of rGO were signified by two prominent peaks in the room temp GA samples' Raman spectra, which are located at 1365 and 1595 cm^{-1} . GA samples that have been annealed at high temperature, on the other hand, show a notable decrease in the strength of the D peak and an extra 2D peak at 2712 cm^{-1} .

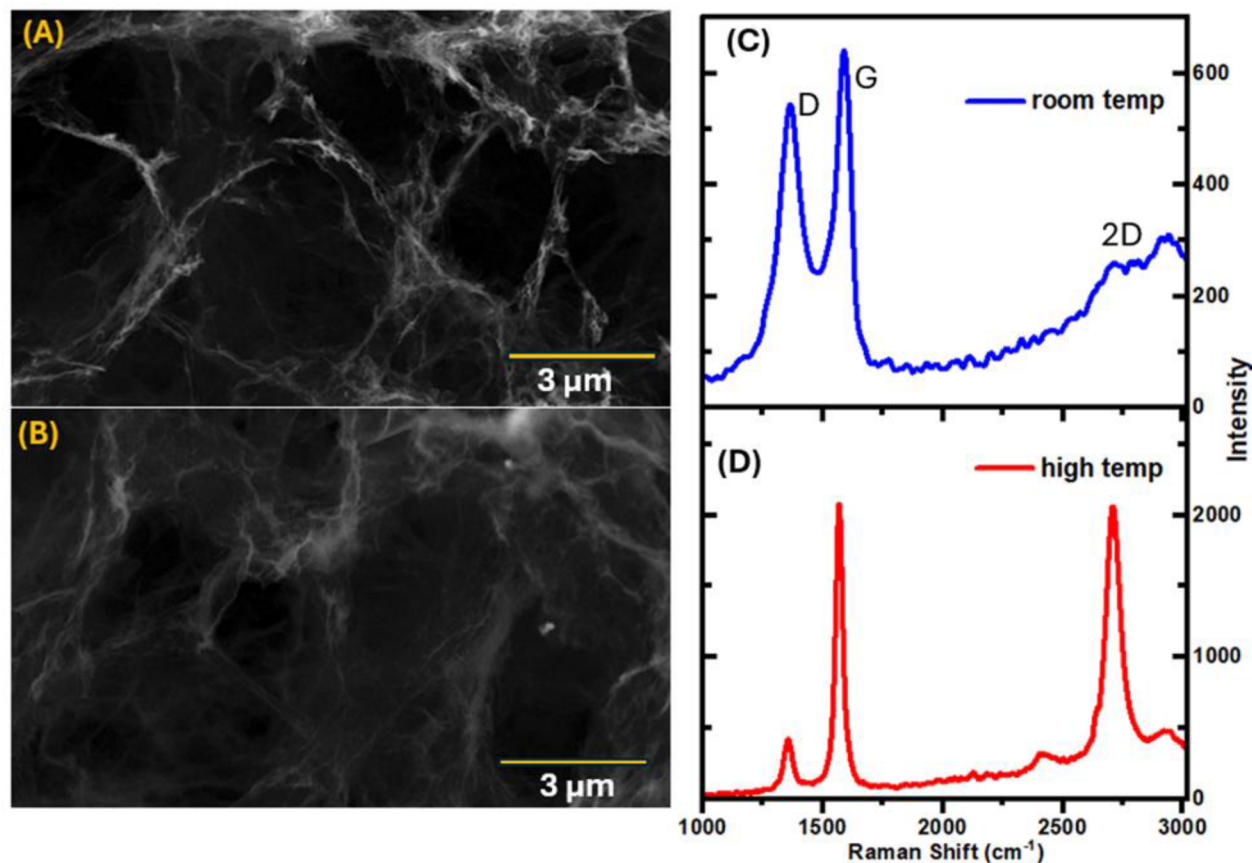


Figure 1—(A, B) Scanning electron micrograph of GA annealed at room temperature and at 2700 °C. (C, D) Raman spectrum of GA annealed at room temperature and at 2700 °C²⁰

As the G band is a characteristic peak of the sp^2 carbon structure and represents the crystallization and symmetry of graphene; the D band is a defect peak and represents the disorders and defects between graphene layers; and the 2D band is a double phonon resonance peak and represents the stack degree of graphene, which is a common symbol to identify the existence of graphene.²² Furthermore, graphene structure flaws can be studied using the intensity ratio of the D–G bands and graphene structure layers can be described using the intensity ratio of the 2D–G bands. The intensity ratio of D–G band is higher for high temperature annealed 3D GA in compared with room temperature synthesised GA. This implies that current annealed 3D graphene show that the graphene is of good quality with few layers and this also illustrates how effectively the annealing enhanced the crystallinity and eliminated oxygen functional groups from the GA, enabling the conversion of rGO to graphene. The pore size also got little bit squeezed can seen in from the SEM image of 1B in compared with SEM image of unannealed graphene shown in Figure 1A. This illustrates how effectively the annealing enhanced the crystallinity and eliminated oxygen functional

groups from the GA, enabling the conversion of reduced graphene oxide to graphene. Because oxygen functional groups are unstable, they desorb at higher annealing temperatures, becoming more oleophilic and hydrophobic. In order to investigate non-binary mechanical forces on a contact and provide better strain and pressure measurements, the previous study used changing contact resistance between a graphene aerogel and a hard metal electrode.²¹ In which, the high-temperature induced covalent cross-linking of graphene in the aerogel provides high strength, durability. Up to 68% tensile strain and 92% compressive strain, it is confirmed this anomalous super elastic behavior in a graphene aerogel composed of covalently cross-linked graphene sheets. The stress-strain behavior of the 3D graphene confirms the elastic bending of graphene sheets and along with higher hydrophobic and oleophilic nature motivates for adsorption experiments for oil-water separation.

Several other groups explore various graphene-based materials for oil spill cleanup and water separation by combining graphene with different suitable systems such as graphene-based membranes; graphene sponges; graphene-decorated meshes; graphene hydrogels; graphene foam; and graphene-coated cotton etc.²³⁻²⁸ This is one of the first attempts to for oil-water separation with 3D Graphene without any other materials involved with graphene. [Figure 2\(a\)](#) shows a schematic image of 3D Graphene used for this experiment. The 3D Graphene is extremely porous and with the hydrophobicity and oleophilicity should provide a good oil/water separation capacity, the porous structure should produce a high absorbance for liquids. A clear ring drop of oil was left placed on top of oil water mixture ([Figure 2B](#)). The 3D graphene swiftly and selectively absorbed the oil from the mixture as shown in [Figure 2C](#). A high-density oil was efficiently absorbed from water by 3D graphene. When a piece of 3D graphene came into contact with oil in water, the oil was quickly absorbed into the pores of the 3D graphene within a few seconds. The experiment's video is included in the supporting information. The absorbed oil collected by squeezing the graphene due to their strong elasticity and again used for next 10 cycles for oil separation. Due to low scale experiment the absorption capacity is not calculated efficiently for this experiment but due to the recoverability, good elasticity nature of 3D Graphene can be used as an effective adsorbent compared with other sponge-based materials. We have proposed a schematic for large scale application to use 3D graphene used as separator membrane for oil-water separation as shown in [Figure 3](#). [Figure 3A](#) propose by use of a large-scale 3D graphene can be prepared for industrial use. The 3D graphene can be utilized as a membrane to separate water from oil after the mixture of water and oil has passed through it. After mechanical pressing the oil contained 3D graphene, oil can be extracted separately, as shown in [Figures 3 B and C](#). This implies that in order to facilitate the transition from reduced graphene oxide to graphene, the annealing process was effective in increasing the crystallinity and eliminating oxygen functional groups from the GA. This indicates that oxygen functional groups are unstable and desorb at higher annealing temperatures, which causes them to become more hydrophobic and oleophilic. When used as an absorption medium to collect oil from water surfaces, the produced 3D graphene showed excellent selectivity due to its high compressive strength, hydrophobicity, and oleophilic. Excellent recyclability is also observed by the fact that 3D graphene's absorption capacity did not degrade and that its weight did not change after ten reuses. This describes potential directions and points of view for future research to better understand the processes involved in oil removal from wastewater. This describes potential directions 3D graphene, with its exceptional capabilities and simple regulation, holds great promise for new pollutant management due to its continuous porous structures. Therefore, it is believed that 3D graphene will become one of the most promising low-cost water purification options. and it is expected that this effort will inspire further future endeavour to advance the use of 3D graphene in wastewater separation and to encourage sustainability.

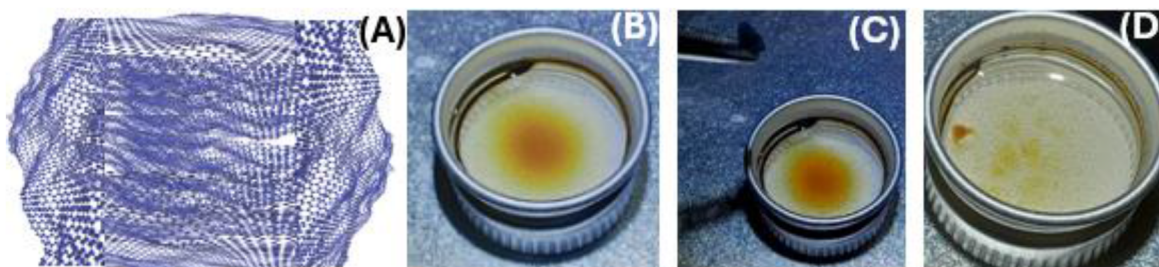


Figure 2—(A) Schematic of 3D Graphene (B) Oil-water mixture (C) adsorption experiment (D) after removal of oil

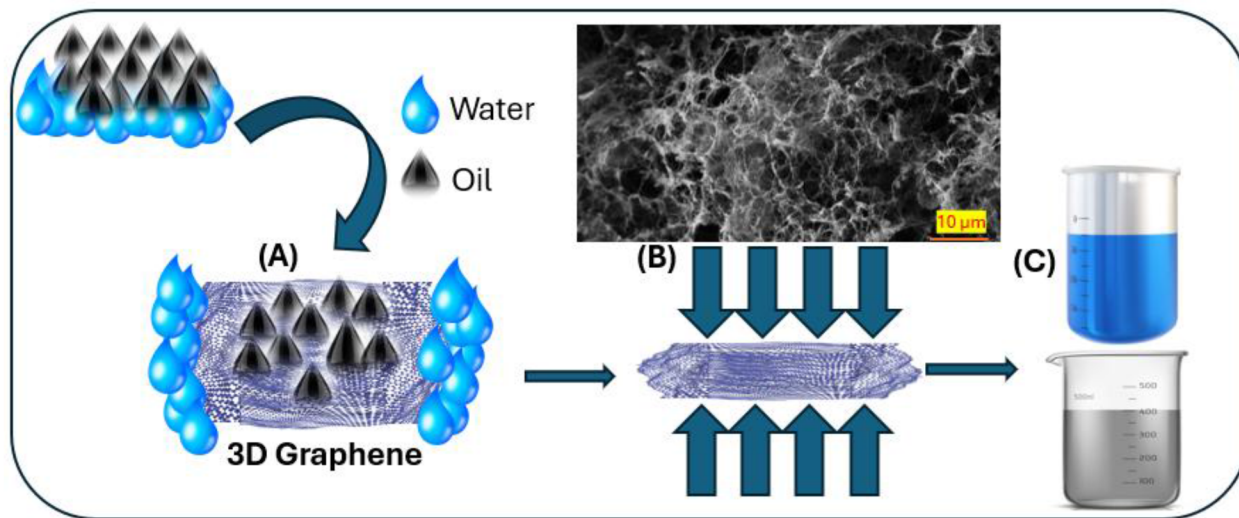


Figure 3—(A) Schematic of 3D Graphene as membrane for oil-water emulsion (B) collection of oil by squeezing (C) Collection of oil and water separately

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References

1. Wong, L. Y.; Lau, S. Y.; Pan, S.; Lam, M. K., 3D graphene-based adsorbents: Synthesis, proportional analysis and potential applications in oil elimination. *Chemosphere* 2022, **287** (Pt 2), 132129.
2. Cheng, C. S.; Deng, J.; Lei, B.; He, A.; Zhang, X.; Ma, L.; Li, S.; Zhao, C., Toward 3D graphene oxide gels based adsorbents for high-efficient water treatment via the promotion of biopolymers. *J Hazard Mater* 2013, **263** Pt 2, 467-78.
3. Hiew, B. Y. Z.; Lee, L. Y.; Lee, X. J.; Thangalazhy-Gopakumar, S.; Gan, S.; Lim, S. S.; Pan, G.-T.; Yang, T. C.-K.; Chiu, W. S.; Khiew, P. S., Review on synthesis of 3D graphene-based configurations and their adsorption performance for hazardous water pollutants. *Process Safety and Environmental Protection* 2018, **116**, 262–286.
4. Liu, H.; Qiu, H., Recent advances of 3D graphene-based adsorbents for sample preparation of water pollutants: A review. *Chemical Engineering Journal* 2020, 393.
5. Purnendu; Satapathi, S., Graphene-Based 3D Xerogel as Adsorbent for Removal of Heavy Metal Ions from Industrial Wastewater. *Journal of Renewable Materials* 2017, **5** (2), 96–102.

6. Şimşek, B.; Sevgili, İ.; Ceran, Ö. B.; Korucu, H.; Şara, O. N., Nanomaterials Based Drinking Water Purification: Comparative Study with a Conventional Water Purification Process. *Periodica Polytechnica Chemical Engineering* 2018, **63** (1), 96–112.
7. Estrada, A. C.; Daniel-da-Silva, A. L.; Leal, C.; Monteiro, C.; Lopes, C. B.; Nogueira, H. I. S.; Lopes, I.; Martins, M. J.; Martins, N. C. T.; Goncalves, N. P. F.; Fateixa, S.; Trindade, T., Colloidal nanomaterials for water quality improvement and monitoring. *Front Chem* 2022, **10**, 1011186.
8. Wang, B.; Feng, S.; Wang, C.; Liu, X.; Chen, L.; Yan, D., Nanostructure-Based Oil–Water Separation: Mechanism and Status. *Separations* 2023, **10** (11).
9. Ghadimi, M.; Zangenehtabar, S.; Homaeigohar, S., An Overview of the Water Remediation Potential of Nanomaterials and Their Ecotoxicological Impacts. *Water* 2020, **12** (4).
10. Ruan, R.; Li, S.; Ji, Y.; Li, X.; Wang, X., Recent advances in novel functional nanomaterials-based oil/water separation techniques. *E3S Web of Conferences* 2021, 272.
11. Winarto; Yamamoto, E.; Yasuoka, K., Separation of water-alcohol mixtures using carbon nanotubes under an electric field. *Phys Chem Chem Phys* 2019, **21** (28), 15431–15438.
12. Brandl, F.; Bertrand, N.; Lima, E. M.; Langer, R., Nanoparticles with photoinduced precipitation for the extraction of pollutants from water and soil. *Nat Commun* 2015, **6**, 7765.
13. Branislav R. S.; Dragana A.; Mića J.; Branimir K.; Jovanović A.; *Removal of mineral oil and wastewater pollutants using hard coal Chemical Industry & Chemical Engineering Quarterly*, 2009, **15** (2) 57–62
14. Choudhury, T. R.; Acter, T.; Uddin, N.; Kamal, M.; Sarwaruddin Chowdhury, A. M.; Safiur Rahman, M., Heavy metals contamination of river water and sediments in the mangrove forest ecosystems in Bangladesh: A consequence of oil spill incident. *Environmental Nanotechnology, Monitoring & Management* 2021, 16.
15. Bonaccorso, F.; Sun, Z.; Hasan, T.; Ferrari, A. C., Graphene photonics and optoelectronics. *Nature Photonics* 2010, **4** (9), 611–622.
16. Wu, J.; Gao, J., Wideband absorption in one dimensional bilayer-graphene embedded photonic multilayer structure. *Superlattices and Microstructures* 2020, 140.
17. Papageorgiou, D. G.; Kinloch, I. A.; Young, R. J., Mechanical properties of graphene and graphene-based nanocomposites. *Progress in Materials Science* 2017, **90**, 75–127.
18. W. Wambu, E., The Graphene Surface Chemistry and Adsorption Science. In *Chemistry of Graphene - Synthesis, Reactivity, Applications and Toxicities*, 2024.
19. Šilhavík, M.; Kumar, P.; Zafar, Z. A.; Mišek, M.; Čičala, M.; Piliarik, M.; Červenka, J., Anomalous elasticity and damping in covalently cross-linked graphene aerogels. *Communications Physics* 2022, **5** (1).
20. Kumar, P.; Silhavik, M.; Parida, M. R.; Nemec, H.; Cervenka, J.; Kuzel, P., Terahertz charge transport dynamics in 3D graphene networks with localization and band regimes. *Nanoscale Adv* 2023, **5** (11), 2933–2940.
21. Kumar, P.; Šilhavík, M.; Zafar, Z. A.; Červenka, J., Contact resistance based tactile sensor using covalently cross-linked graphene aerogels. *Nanoscale* 2022, **14** (4), 1440–1451.
22. Ferrari, A. C.; Basko, D. M., Raman spectroscopy as a versatile tool for studying the properties of graphene. *Nat Nanotechnol* 2013, **8** (4), 235–46.
23. Dattatray A.; Pravin H. W.; Rahul V. K.; Surajit S., Super-hydrophobic carrageenan cross-linked graphene sponge for recovery of oil and organic solvent from their water mixtures. *Polymer Testing* 90 2020, **90**, 106743

24. Azam J.; Elaheh K.G.; Flame retardant, superhydrophobic, and superoleophilic reduced grapheneoxide/orthoaminophenolpolyurethanespongeforefficientoil/ water separation. *Journal of Molecular Liquids* 2020, **307** 112979
25. Tiefan H.; Lin Z.; Huanlin C.; Congjie G.; Sol–gel fabrication of a non-laminated graphene oxide membrane for oil/water separation. *Mater. Chem. A*, 2015, **3**, 19517–19524
26. Bong, J.; Lim, T.; Seo, K.; Kwon, C. A.; Park, J. H.; Kwak, S. K.; Ju, S., Dynamic graphene filters for selective gas-water-oil separation. *Sci Rep* 2015, **5**, 14321.
27. Shiu, R. F.; Lee, C. L.; Hsieh, P. Y.; Chen, C. S.; Kang, Y. Y.; Chin, W. C.; Tai, N. H., Superhydrophobic graphene-based sponge as a novel sorbent for crude oil removal under various environmental conditions. *Chemosphere* 2018, **207**, 110–117.
28. Singhal, A. V.; George, R.; Sharma, A. K.; Malwal, D.; Lahiri, I., Development of superhydrophilic tannic acid-crosslinked graphene oxide membranes for efficient treatment of oil contaminated water with enhanced stability. *Heliyon* 2020, **6** (10), e05127.